# Open Science

## Today's agenda:

- Final presentation reminder
- Code and Data sharing
- Coding time in class (optional)

# Final Presentations

- 12:30-2:30 p.m. on Tuesday, December 9th
- Remember to have your slides added to the Google Slides file beforehand
- Practice your timing to make sure you're under the 5 minutes

# What is Open Science?

Open Science: “transparent and accessible knowledge that is shared and developed through collaborative networks”

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Gallagher, *et al.* (2020) *NEE*

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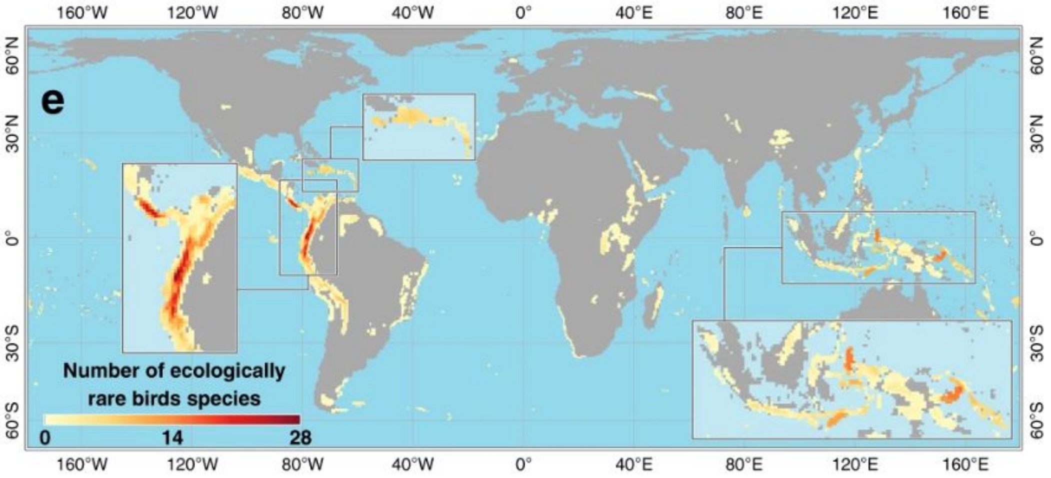
# What are Open Data?

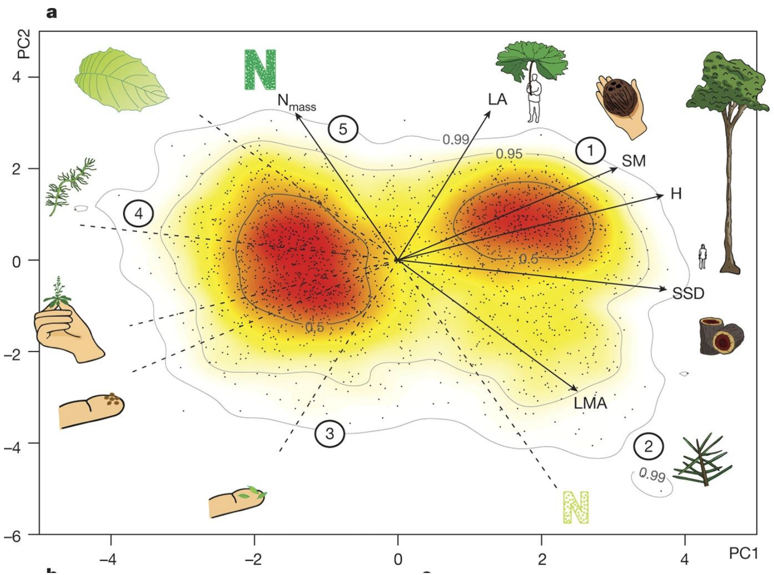
- **Available and Accessible:** the data must be ***available as a whole*** and at no more than a reasonable reproduction cost, preferably by downloading over the internet. The data must also be available in a ***convenient and modifiable form***.

- **Reusable and Redistriable:** the data must be provided under terms that ***permit re-use and redistribution*** including the intermixing with other datasets.

- **Universal Participation:** ***everyone must be able to use***, re-use and redistribute - there should be no discrimination against fields of endeavour or against persons or groups.

# Global Questions Need Global Participation

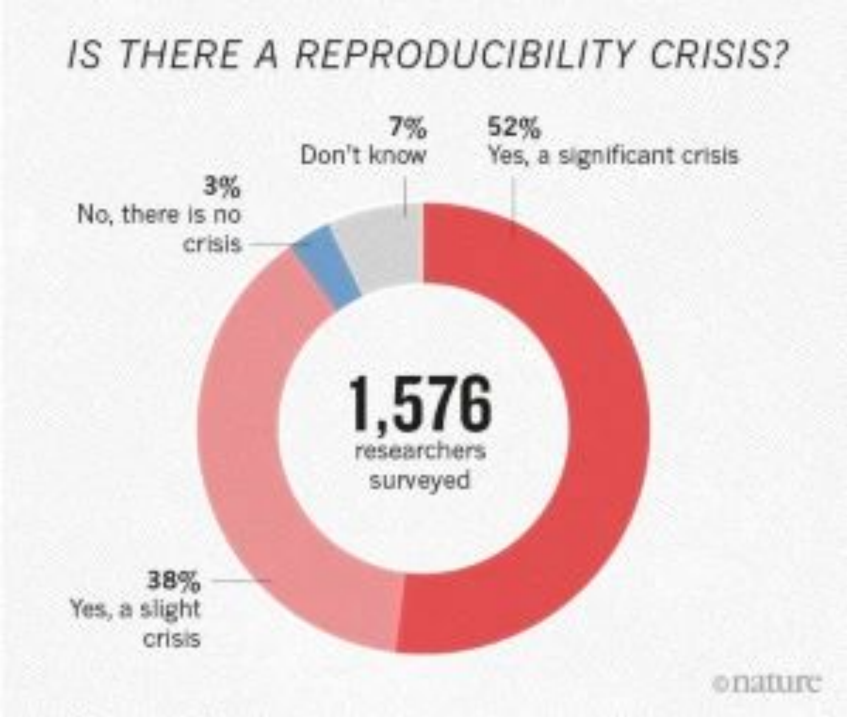



Global spectrum of plant form and function

Loiseau, et al. (2020) *Nature Comm;* Diaz, *et al.* (2016) Nature.

# Open Data Enhance Reproducibility



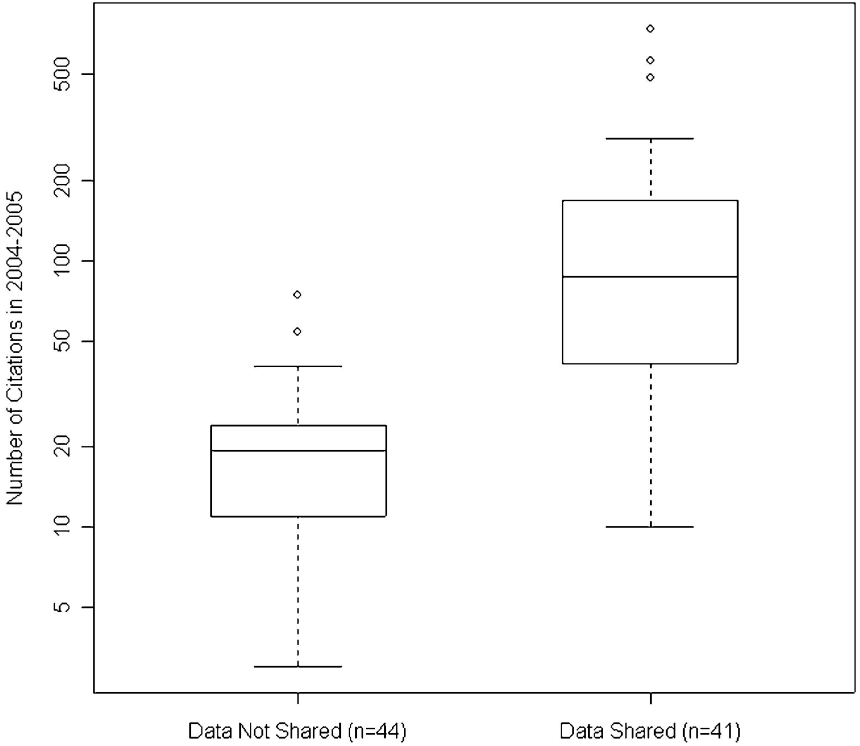

Baker. (2016) Nature.

# Ethical Responsibilities

- **Publicly funded data are public property**
- **Transparency is a key part of science**
- **Open Data increase inclusion**
- **Open Data save money!**

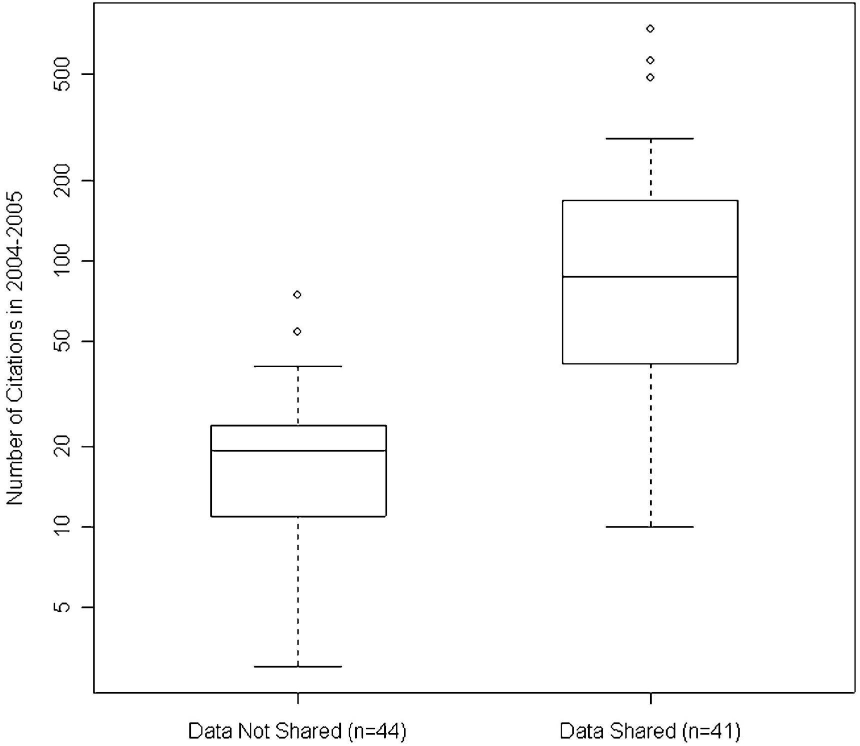
Piwowar, et al. (2011) *Nature*

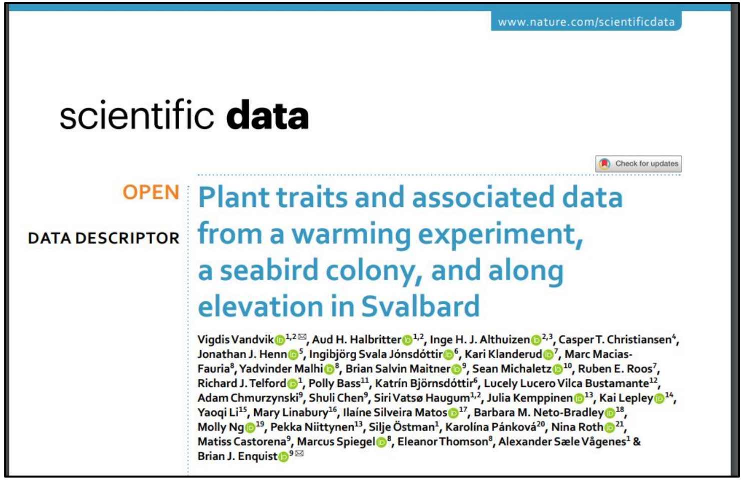
# Open Data Improve Citations




Piwowar, *et al.* (2007) *PLOSONE*

# Open Data Improve Citations and Publications





www.nature.com/scientificdata

scientific **data**

Check for updates

OPEN

DATA DESCRIPTOR

## Plant traits and associated data from a warming experiment, a seabird colony, and along elevation in Svalbard

Vigdis Vandvik[1,2], Aud H. Halbritter[1,2], Inge H. J. Althuizen[2,3], Casper T. Christiansen[4], Jonathan J. Henn[5], Ingibjörg Svala Jónsdóttir[6], Kari Klanderud[7], Marc Macias-Fauria[8], Yadvinder Malhi[8], Brian Salvin Maitner[9], Sean Michaletz[10], Ruben E. Roos[7], Richard J. Telford[1], Polly Bass[11], Katrín Björnsdóttir[6], Lucely Lucero Vilca Bustamante[12], Adam Chmurzynski[9], Shuli Chen[9], Siri Vatsø Haugum[1,2], Julia Kemppinen[13], Kai Lepley[14], Yaoqi Li[15], Mary Linabury[16], Ilaíne Silveira Matos[17], Barbara M. Neto-Bradley[18], Molly Ng[19], Pekka Niittynen[13], Silje Östman[1], Karolína Pánková[20], Nina Roth[21], Matiss Castorena[9], Marcus Spiegel[8], Eleanor Thomson[8], Alexander Sæle Vågenes[1] & Brian J. Enquist[9]

Piwowar, *et al.* (2007) *PLOSONE*

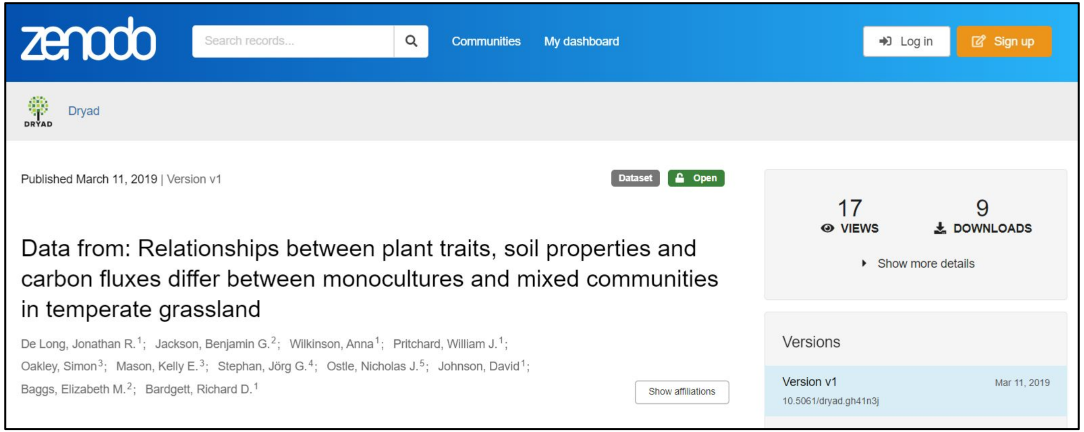
# How to share data

**Range of options**

Keller, *et al.* (2022) *MEE*

# How to share data

**Range of options**




Keller, *et al.* (2022) *MEE*

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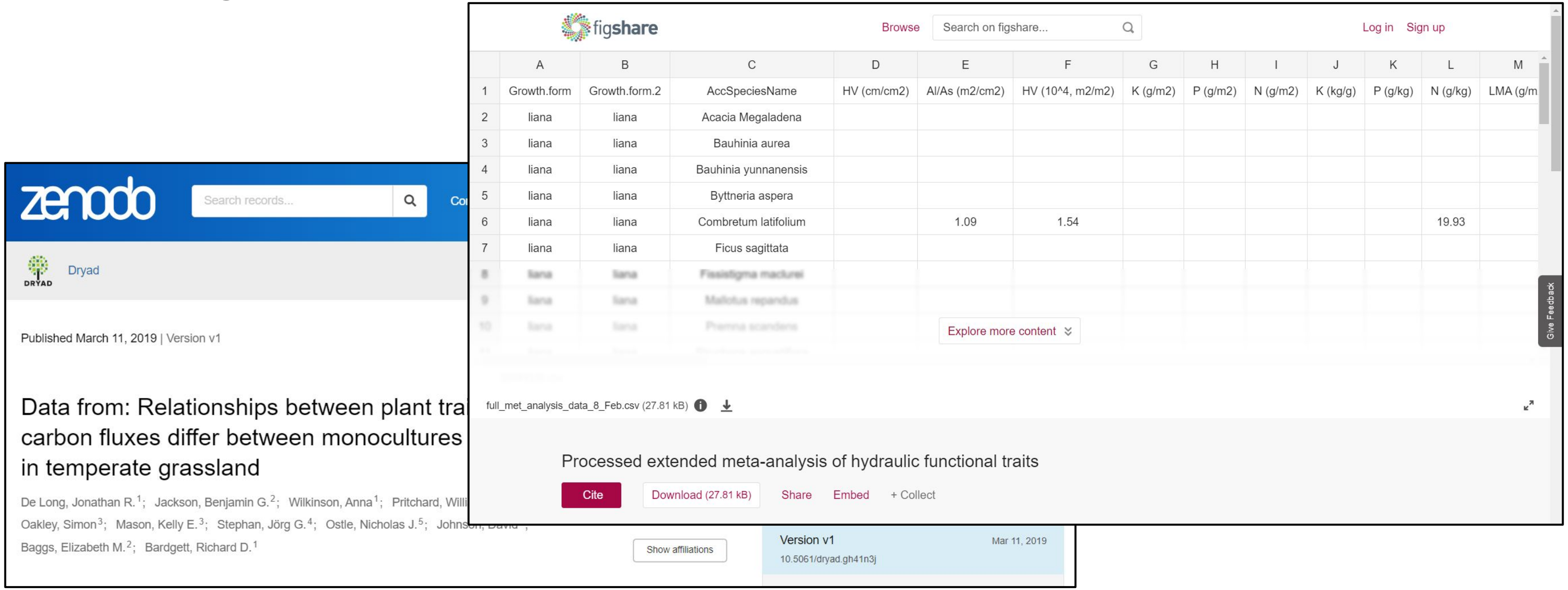
**Range of options**




Keller, *et al.* (2022) *MEE*

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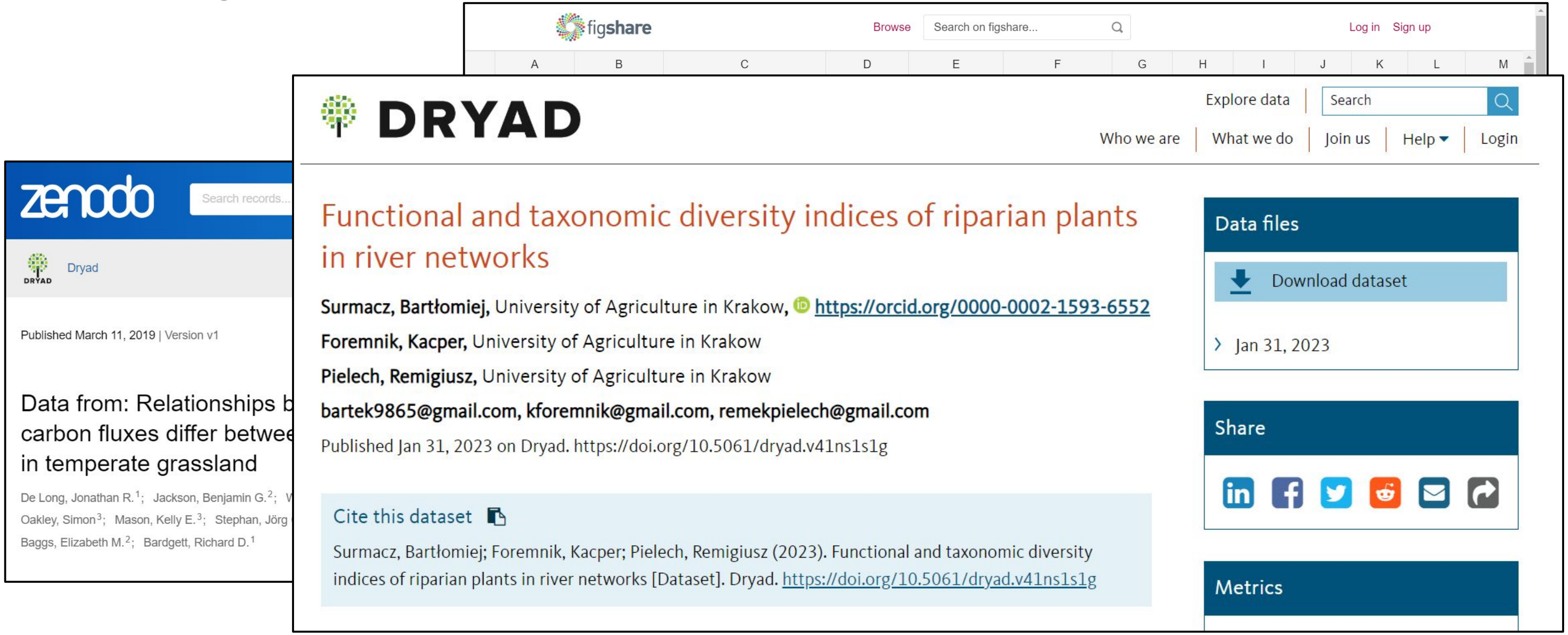


Keller, *et al.* (2022) *MEE*

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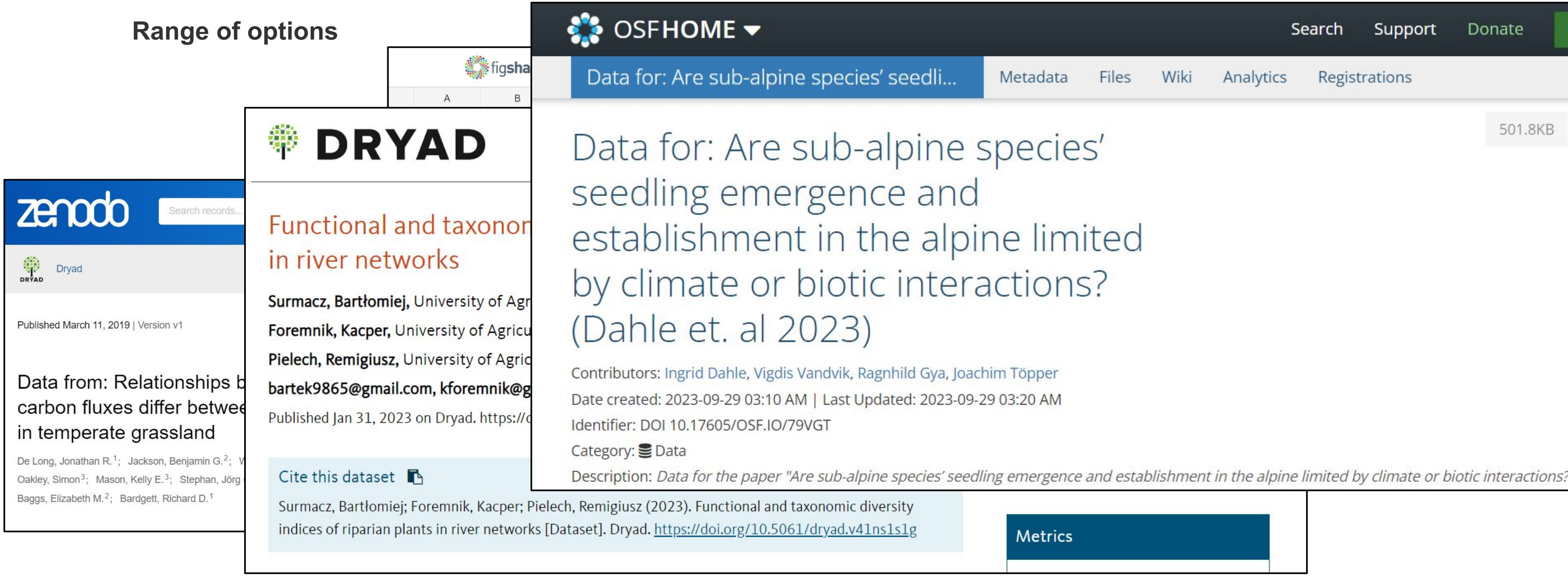
**Range of options**

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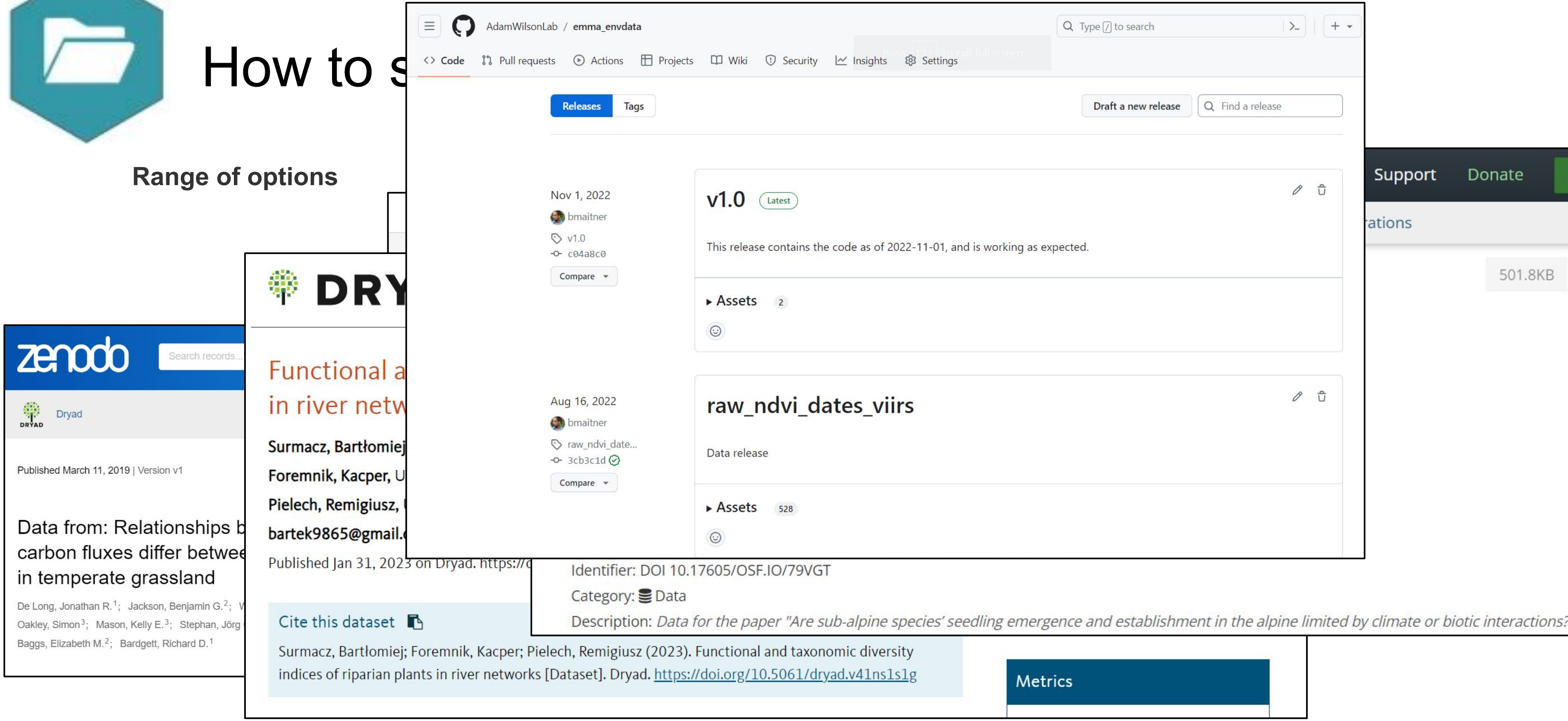


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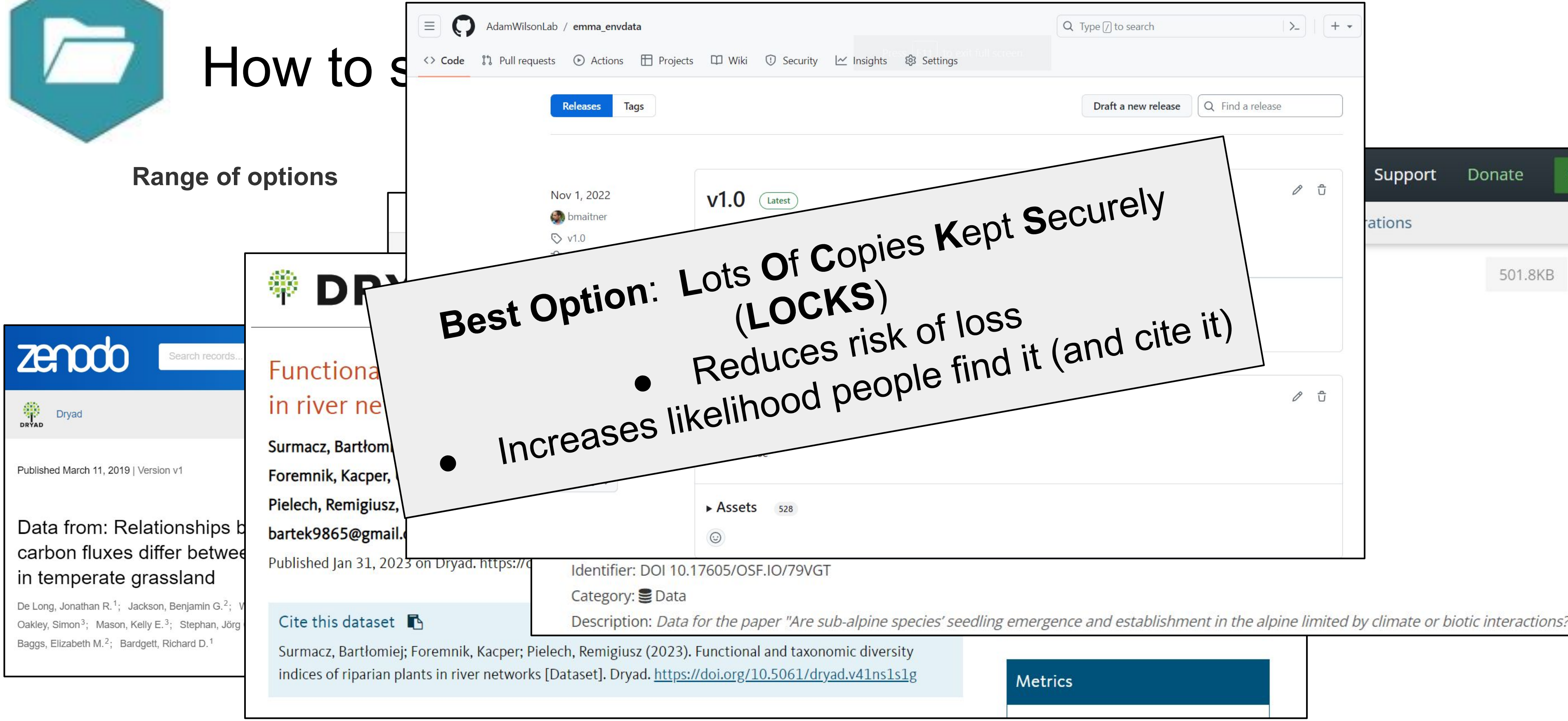


**Best Option**: **L**ots **O**f **C**opies **K**ept **S**ecurely (**LOCKS**)

- Reduces risk of loss
- Increases likelihood people find it (and cite it)

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https://opensource.org/osd/

# What is Open Source?

- **Free Redistribution** The license shall not restrict any party from selling or giving away the software as a component of an aggregate software distribution containing programs from several different sources. The license ***shall not require a royalty or other fee*** for such sale.

- **No Discrimination Against Persons or Groups** The license must not discriminate against any person or group of persons.

- **No Discrimination Against Fields of Endeavor** The license must not restrict anyone from making use of the program in a specific field of endeavor. For example, it may not restrict the program from being used in a business, or from being used for genetic research.

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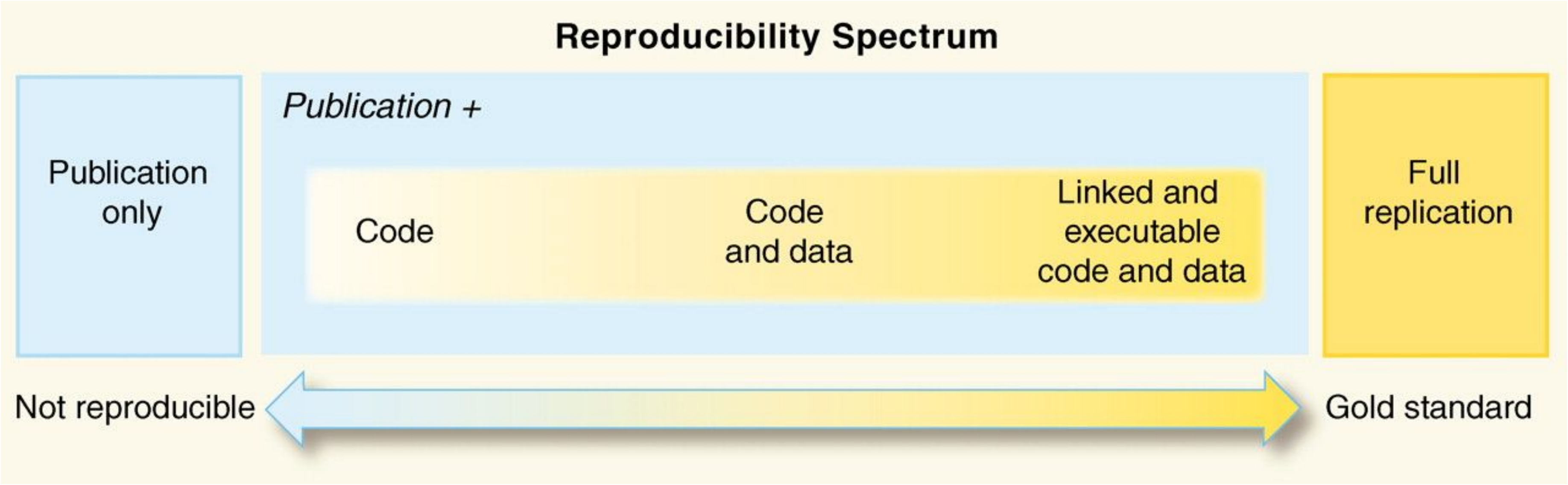
- **Code that is free to use, both in terms of costs and restrictions**

https://opensource.org/osd/

# Reproducibility and Transparency




Peng. (2011) *Science*

# Open Source Reduces Wasted Efforts

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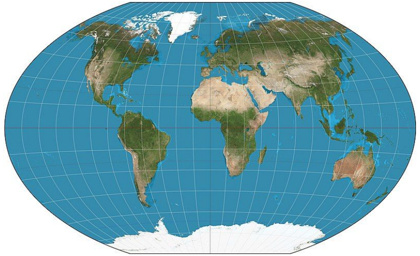

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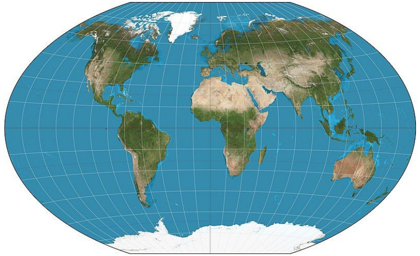
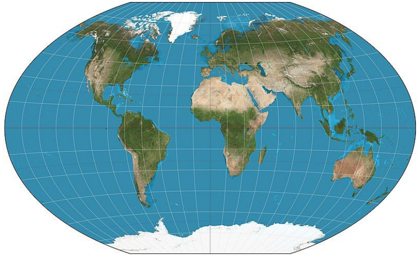



Data

Data

# Open Source Reduces Wasted Efforts



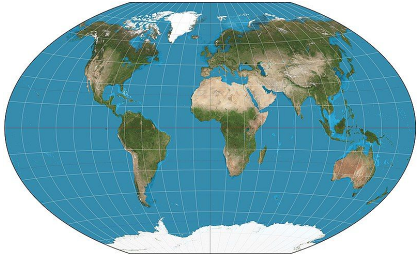
Data

Data

Integration

~ 1 week

# Open Source Reduces Wasted Efforts



Data → Integration → Cleaning

Data → Integration

~ 1 week

~ 2 weeks

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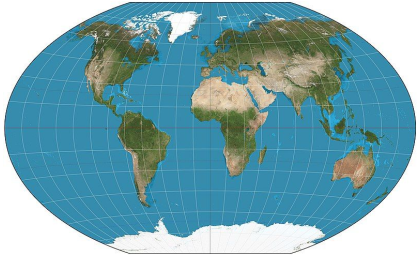

Data

Data

Integration

Cleaning

Analyses

~ 1 week

~ 2 weeks

~ 4 weeks

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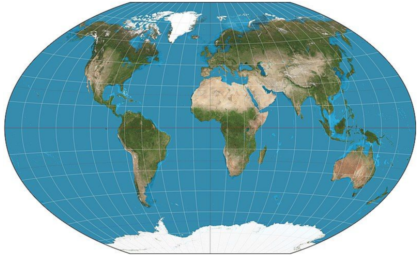


Data → Integration
Data → Integration

Integration → Cleaning → Analyses → Figures

~ 1 week

~ 2 weeks

~ 4 weeks

~ 2 weeks

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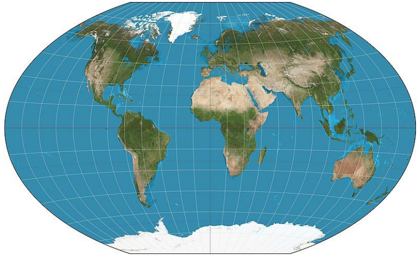


~ 10 minutes

Publish Code

Data

Data

Integration

Cleaning

Analyses

Figures

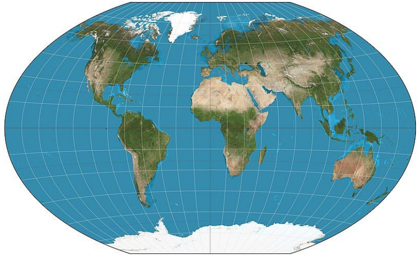
~ 1 week

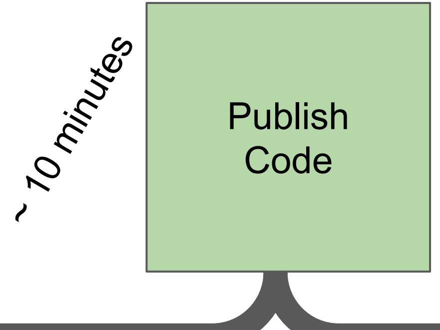
~ 2 weeks

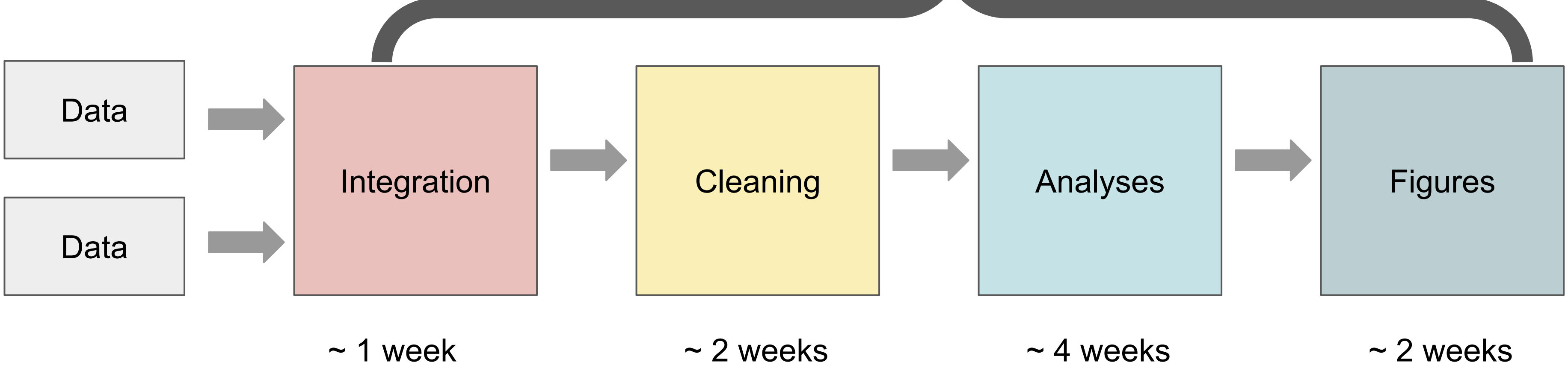
~ 4 weeks

~ 2 weeks

# Open Source Reduces Wasted Efforts





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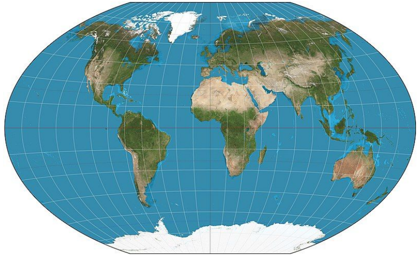

~ 10 minutes

Publish Code

Data

Data

Integration → Cleaning → Analyses → Figures

~ 1 week

~ 2 weeks

~ 4 weeks

~ 2 weeks

Macaw: Andrew Pons, Ctenosaur: Dario

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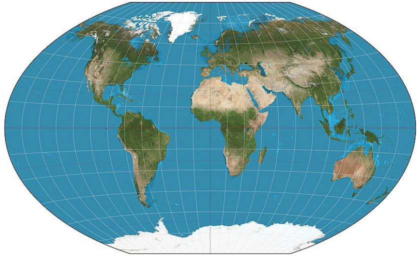

~ 10 minutes

Publish Code

**I saved people months of work!**
(and got a few good citations)

Data

Data

Integra

Figures

~ 1 week

~ 2 weeks

~ 4 weeks

~ 2 weeks

Macaw: Andrew Pons, Ctenosaur: Dario

# Open Source Reduces Barriers

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**ArcGIS Pro All Extensions Bundle** Requires foundational user type

- Esri's most popular ArcGIS extensions bundled together
- Comprehensive set of advanced GIS tools

Qty − 1 +

$1,950/yr

Add to cart

Hide description

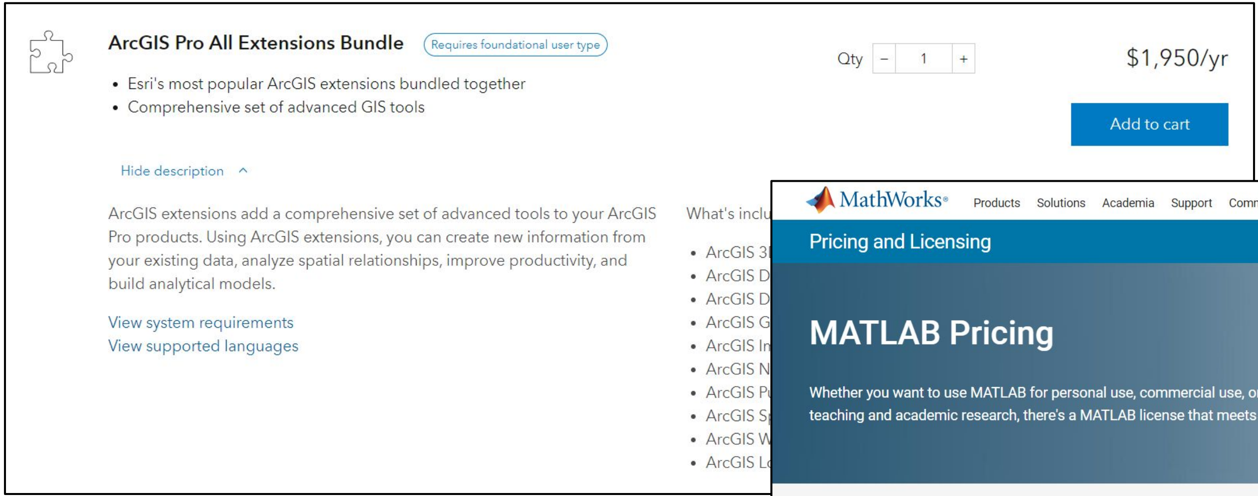
ArcGIS extensions add a comprehensive set of advanced tools to your ArcGIS Pro products. Using ArcGIS extensions, you can create new information from your existing data, analyze spatial relationships, improve productivity, and build analytical models.

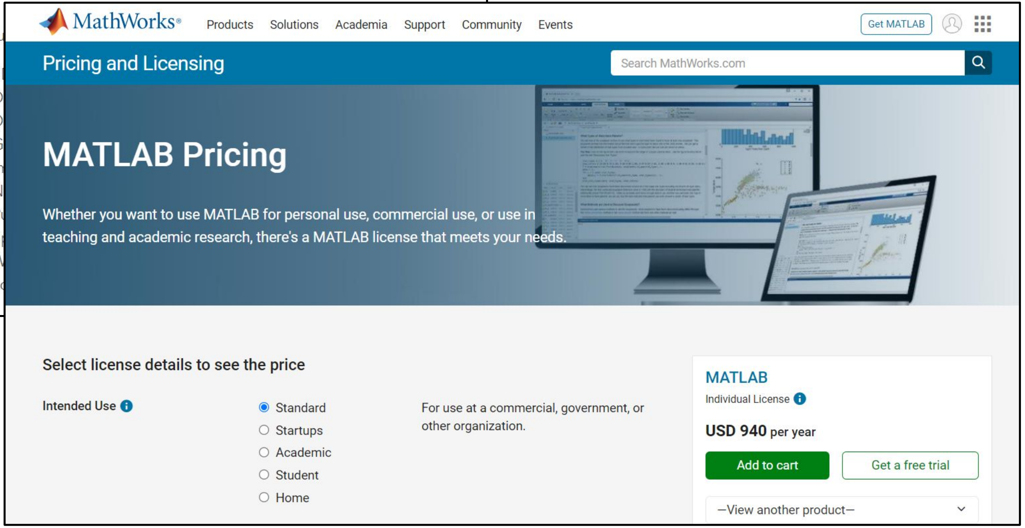
View system requirements
View supported languages

What's included:

- ArcGIS 3D Analyst
- ArcGIS Data Interoperability
- ArcGIS Data Reviewer
- ArcGIS Geostatistical Analyst
- ArcGIS Image Analyst
- ArcGIS Network Analyst
- ArcGIS Publisher
- ArcGIS Spatial Analyst
- ArcGIS Workflow Manager
- ArcGIS LocateXT

# Open Source Reduces Barriers



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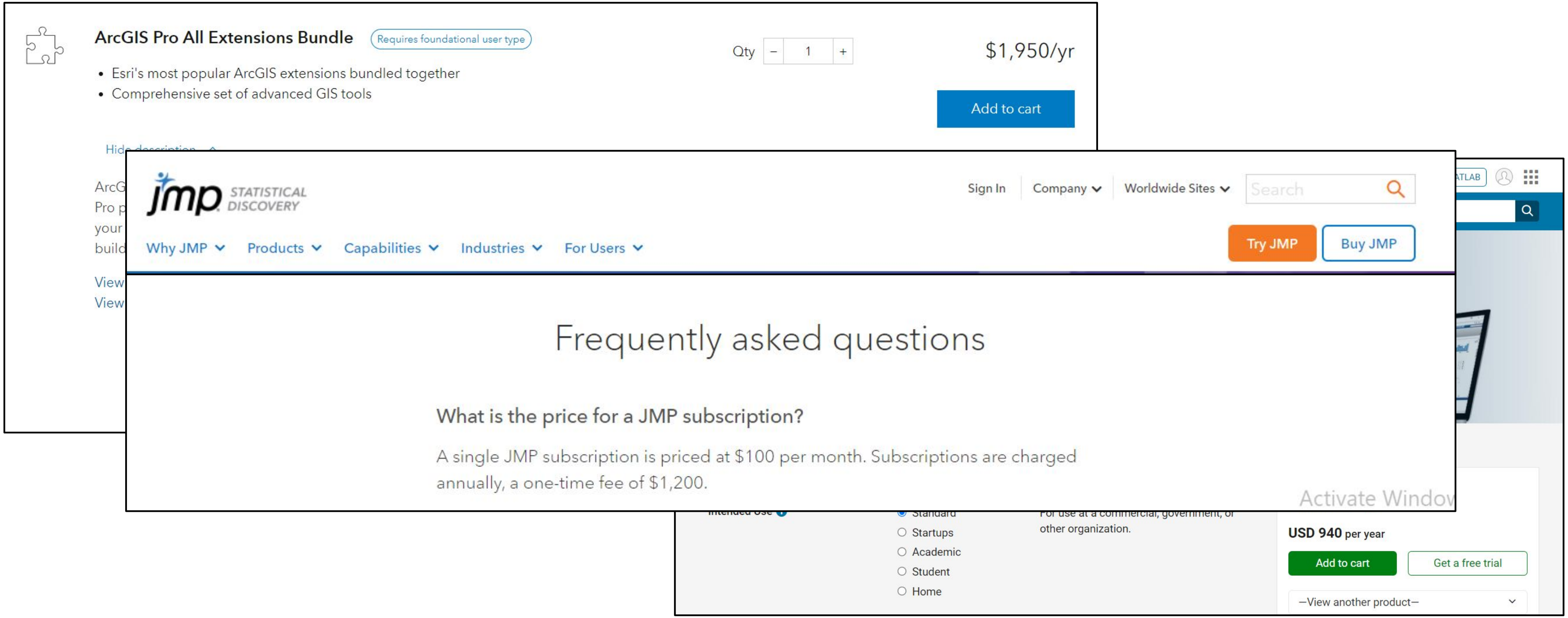
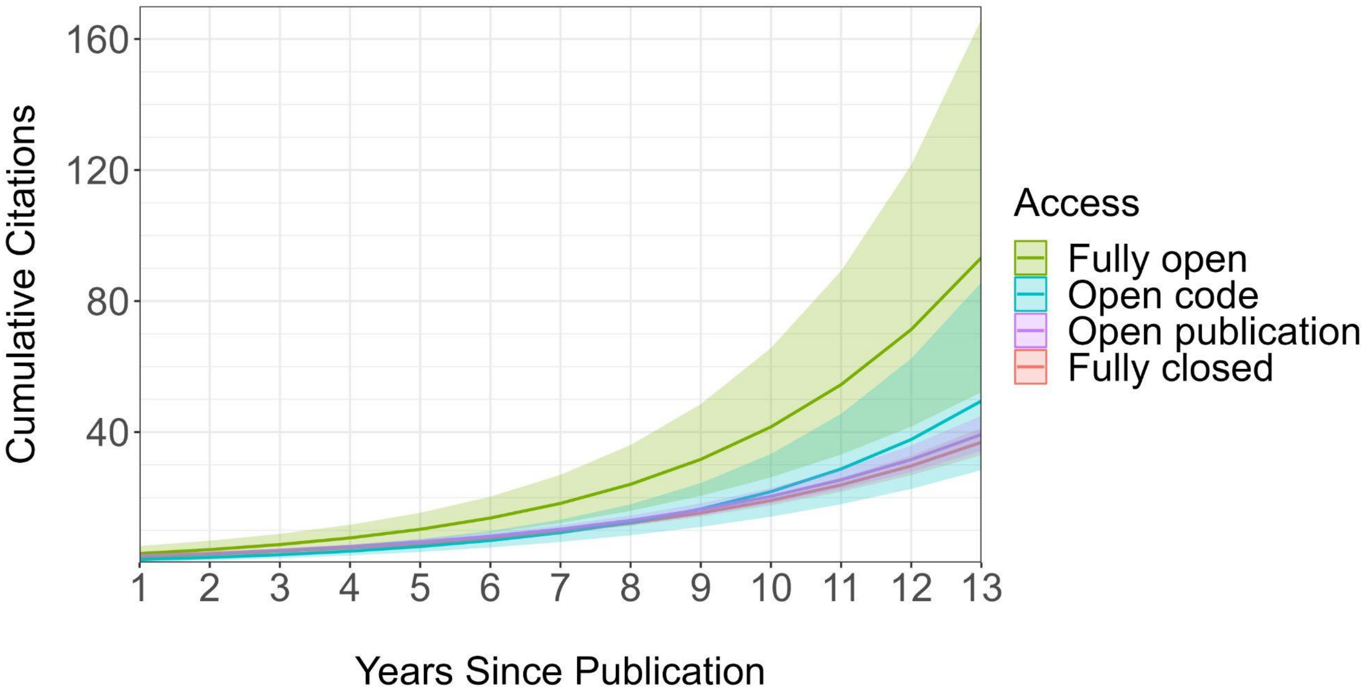


# Open Source Reduces Barriers

# Code Sharing Increases Citations




Maitner, et al. (*in revision*)

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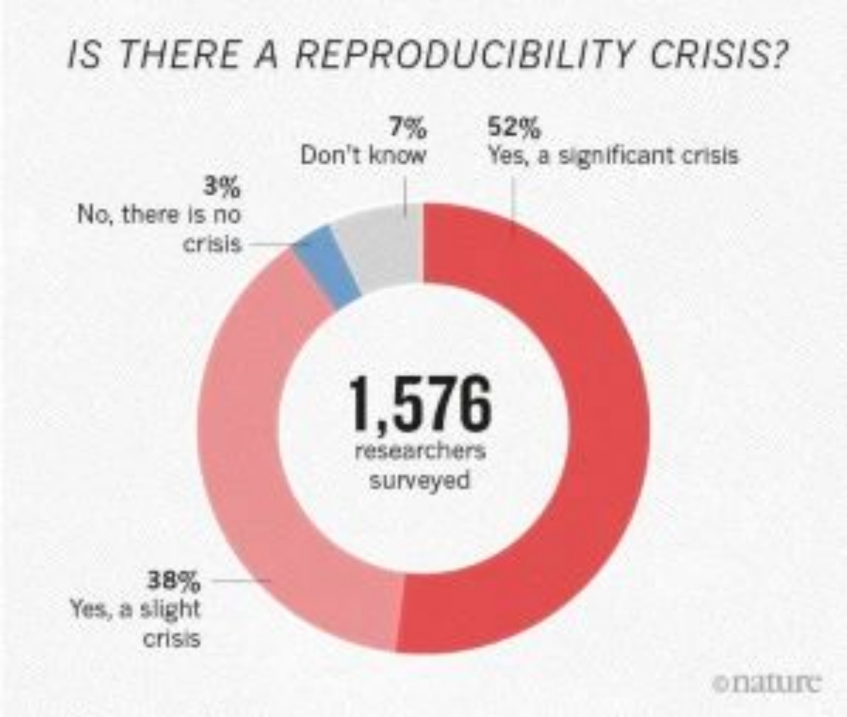
# What are Open Methods?

- Making methods underlying research open and freely accessible

  - **Examples:**
    - Laboratory protocols
    - Data collection protocols
    - Metadata standards
    - Materials
    - Reagents
    - Code (Overlap with Open Source)

# Open Methods Increase Transparency



# Shared Methods Increase Confidence

**CSIRO** PUBLISHING

*Australian Journal of Botany*, 2013, **61**, 167–234
http://dx.doi.org/10.1071/BT12225

**New handbook for standardised measurement of plant functional traits worldwide**

*N. Pérez-Harguindeguy[A,Y], S. Díaz[A], E. Garnier[B], S. Lavorel[C], H. Poorter[D], P. Jaureguiberry[A], M. S. Bret-Harte[E], W. K. Cornwell[F], J. M. Craine[G], D. E. Gurvich[A], C. Urcelay[A], E. J. Veneklaas[H], P. B. Reich[I], L. Poorter[J], I. J. Wright[K], P. Ray[L], L. Enrico[A], J. G. Pausas[M], A. C. de Vos[F], N. Buchmann[N], G. Funes[A], F. Quétier[A,C], J. G. Hodgson[O], K. Thompson[P], H. D. Morgan[Q], H. ter Steege[R], M. G. A. van der Heijden[S], L. Sack[T], B. Blonder[U], P. Poschlod[V], M. V. Vaieretti[A], G. Conti[A], A. C. Staver[W], S. Aquino[X] and J. H. C. Cornelissen[F]*

Received: 31 May 2019 | Accepted: 28 October 2019

DOI: 10.1111/2041-210X.13331

RESEARCH ARTICLE

Methods in Ecology and Evolution BRITISH ECOLOGICAL SOCIETY

**The handbook for standardized field and laboratory measurements in terrestrial climate change experiments and observational studies (ClimEx)**

Aud H. Halbritter[1] | Hans J. De Boeck[2] | Amy E. Eycott[3,4] | Sabine Reinsch[5] | David A. Robinson[5] | Sara Vicca[2] | Bernd Berauer[6] | Casper T. Christiansen[7] | Marc Estiarte[8,9] | José M. Grünzweig[10] | Ragnhild Gya[1] | Karin Hansen[11,12] | Anke Jentsch[6] | Hanna Lee[7] | Sune Linder[13] | John Marshall[14] | Josep Peñuelas[8,9] | Inger Kappel Schmidt[15] | Ellen Stuart-Haëntjens[16] | Peter Wilfahrt[6] | the ClimMani Working Group* | Vigdis Vandvik[1]

# Shared Methods Increase Compatibility

- Data Integration is a major time sink
- Using shared metadata standards and methods increases interoperability

REVIEW

Methods in Ecology and Evolution BRITISH ECOLOGICAL SOCIETY

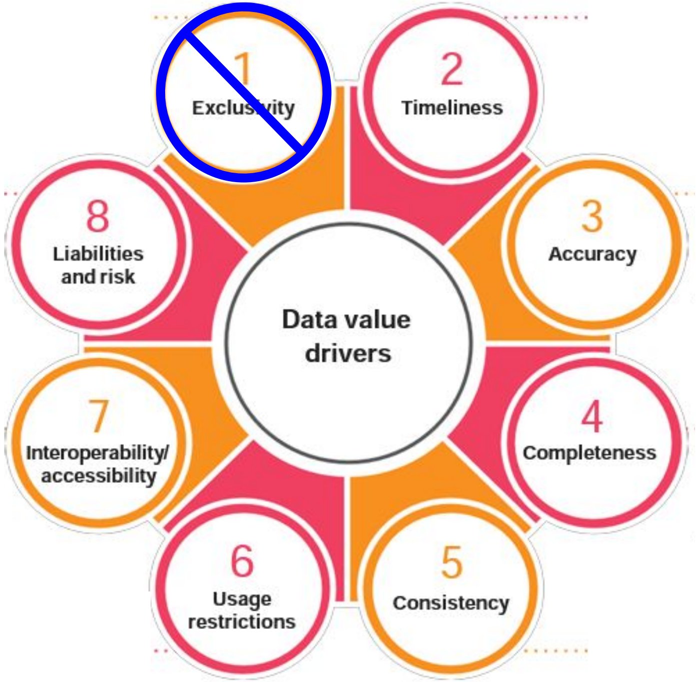
## Towards an ecological trait-data standard

Florian D. Schneider[1] | David Fichtmueller[2] | Martin M. Gossner[3] | Anton Güntsch[2] | Malte Jochum[4,5,6] | Birgitta König-Ries[7] | Gaëtane Le Provost[8] | Peter Manning[8] | Andreas Ostrowski[7] | Caterina Penone[4] | Nadja K. Simons[9,10]

# Open Methods Increase Data Value


https://www.pwc.co.uk/data-analytics/documents/putting-value-on-data.pdf

# Open Methods Increase Data Value

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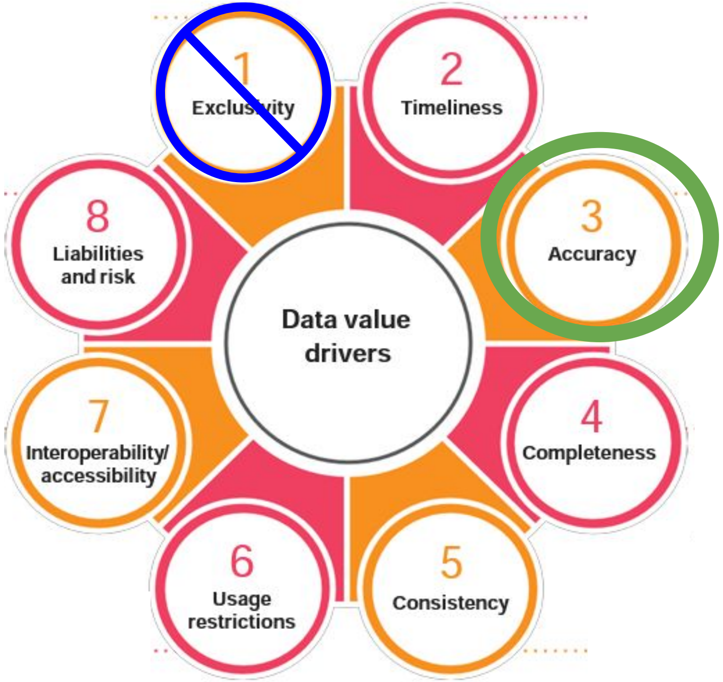


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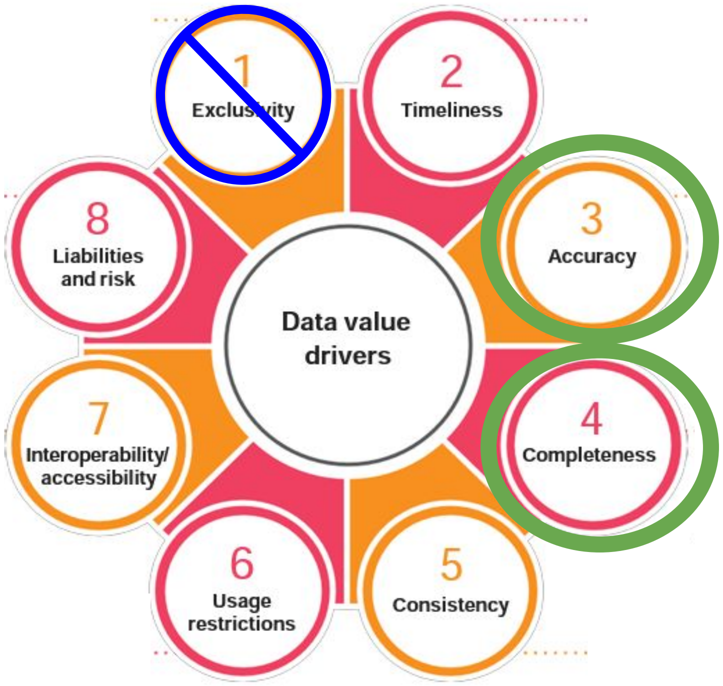


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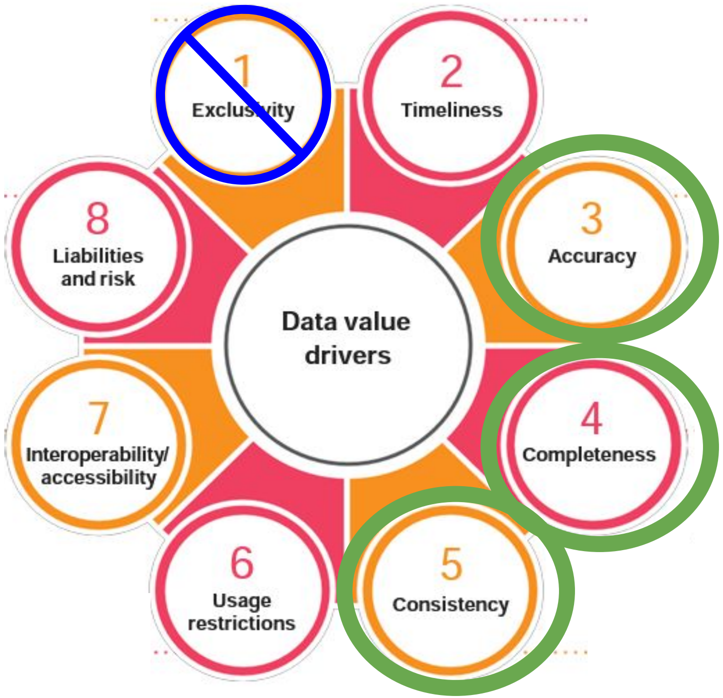


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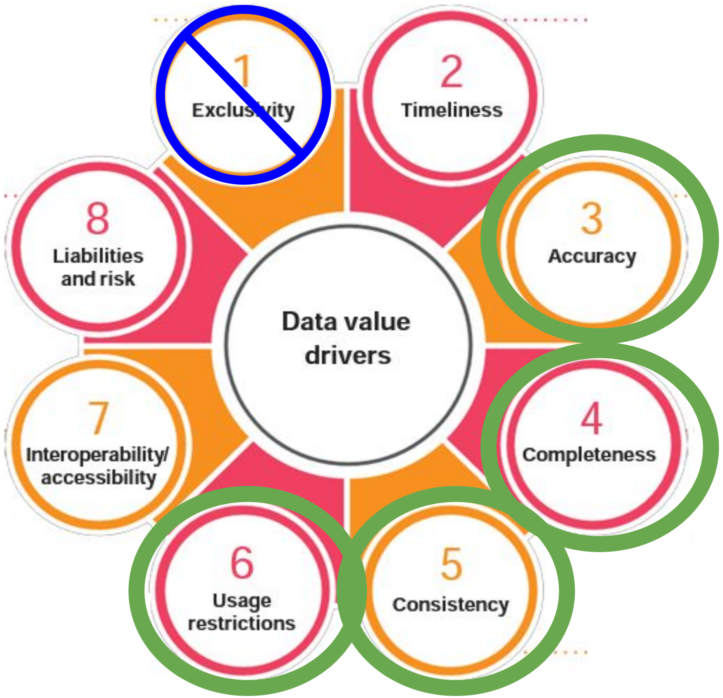


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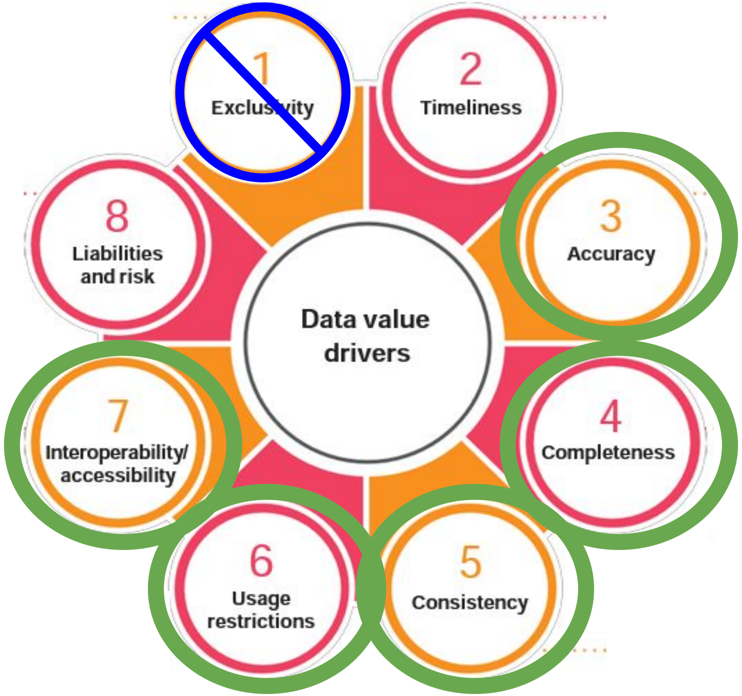


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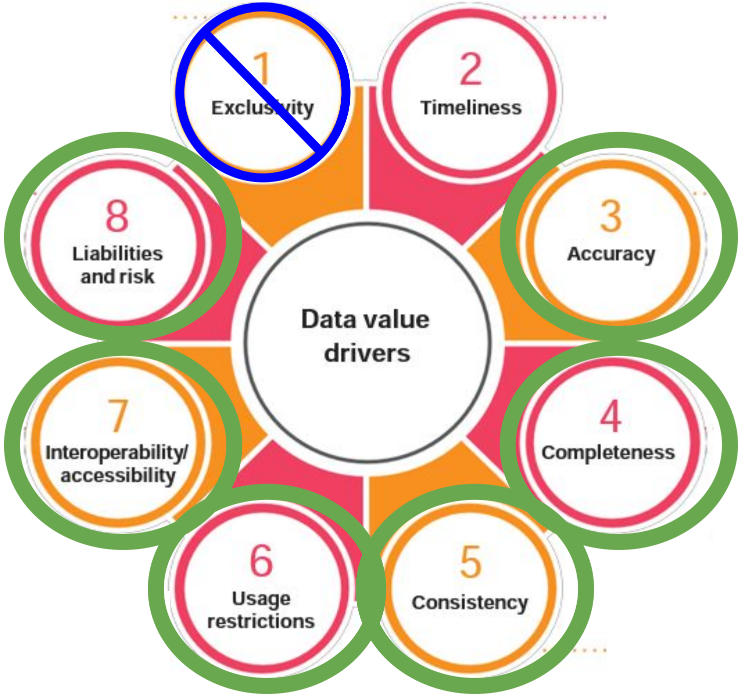


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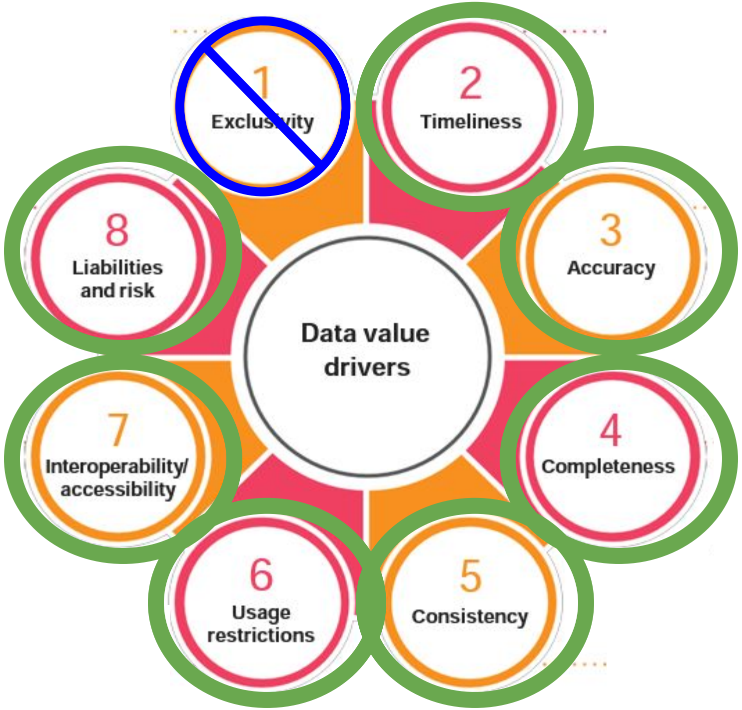
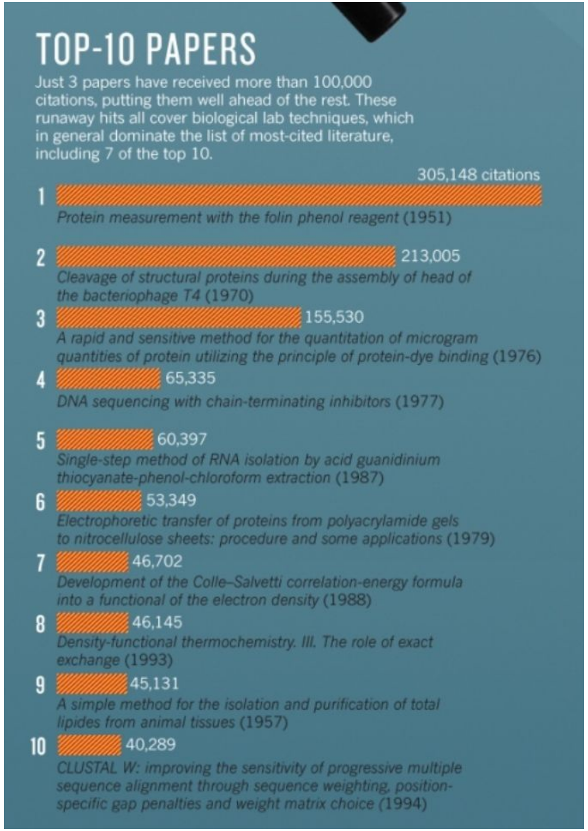



https://www.pwc.co.uk/data-analytics/documents/putting-value-on-data.pdf

# Methods Papers Are Highly Cited




Van Noorden, et al. (2014) Nature

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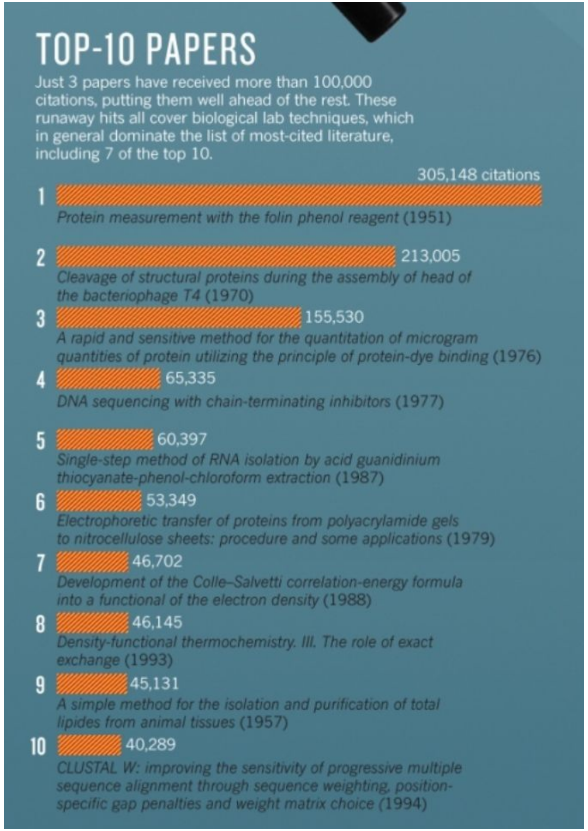
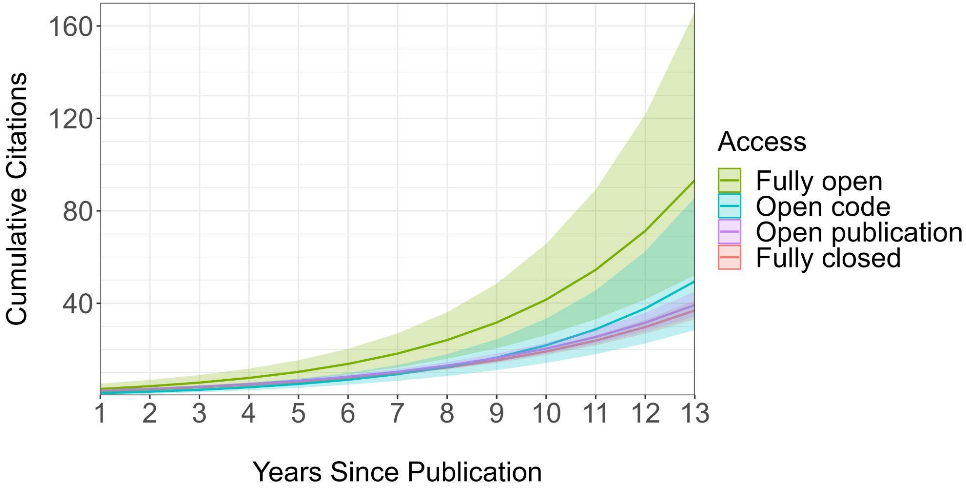



(as are open access papers, so presumably Open Methods will be especially successful? )

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# What is Open Science?

Open Science: “transparent and accessible knowledge that is shared and developed through collaborative networks”


Gallagher, *et al.* (2020) *NEE*

**Before next class:**

- Prepare your presentations